

What a domain prior can and cannot replace

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TGDC case study, research track

A growing number of companies sell generated enterprise data, and they build it from a description of a business rather than from anyone's real records: an org chart, a list of the work the business does, and hand-written rules for how that work behaves. The appeal is obvious, because a client who cannot share their data can still be served. The open question is what that costs. Distributional fidelity is easy to measure and is already known to be a poor guide to whether the data is useful downstream, so the more practical question is which properties of real records the hand-written rules can actually supply, and which ones a client has to hand over.

This study uses public 311 service request feeds from four cities. Each ticket is a unit of work, and the question asked of it is how long it takes to close. A city's service catalogue lists job types and the departments that own them with no timings attached, so it stands in for the description of a business, and the recorded durations stand in for the client records a generator does without.

Hypothesis

My hypothesis is that rules reasoned from the nature of the work, written without sight of any outcome, predict how long a job takes at an AUC between 0.60 and 0.72, and that anything below 0.55 would mean the reasoning adds nothing.

Method

The records are written by the people who do the work. Each feed gives the work type, the owning department, how the request came in, and when it opened and closed. I take two full calendar weeks from each city, the weeks of 4 May and 8 June 2026.

Tickets still open when I pulled the data are kept and counted as slow, since all of them are months past any threshold I use. Chicago closes most of its feed within a second of creation, and I drop those rows because they are records rather than work.

The task is to predict, at the moment a ticket opens, whether it will take longer than the median to close. Recovered skill is the share of the achievable margin over chance that an arm reaches.

I fixed that band before running anything. Recovered skill, the unit I report alongside AUC, is $(\text{AUC} - 0.5) / (\text{ceiling} - 0.5)$. The null I test against is that the rules carry no information about which kind of work runs slow, so that any score they reach is available from carving up the same taxonomy at random.

I need a ceiling to measure against. A prior reads two columns, the service category and the owning department, so I fit the conditional rate from those same two columns to see what they support. This is a taxonomy ceiling, not a general limit on skill, and it separates a weak rule from

weak features. It is one estimator rather than a proved upper bound, so a rule can beat it, and one does. I settled on it after the first Austin arm had already been scored against a different estimator, so the AUCs here are pre-registered and the denominator is not. Every percentage is reported beside its raw AUC for that reason.

I report two kinds of interval, because a prior is a function of the category and gives the same answer to every ticket in one. Resampling tickets treats 17,007 San Francisco rows as independent when I really have 37, so I resample both ways [1]. That second unit needs a warning. Its consistency results are asymptotic in the number of clusters and assume rough balance, and mine is neither: one San Francisco category holds 34% of the rows, and the effective count is 5.3 against 37 categories, Austin 22.3 against 117. At that size almost nothing could clear the interval, so I read the category intervals as descriptive rather than as tests, and where I say something is not established I mean this design cannot establish it [12].

I pulled each catalogue with the query limited to category names and row counts, so no duration, rate or outcome reached me. I wrote the rules from that alone, committed them with a prediction, and only then evaluated. I amended one San Francisco file 27 seconds after committing it, to match on word boundaries rather than substrings, before that city's data existed on disk.

The biggest risk is that the result describes one person's reasoning, so Austin and San Francisco each have two sets of rules under the same blind condition. I wrote one and a language model wrote the other from the same catalogue. New York is the contrast, written while I could see the real rates, and Chicago is where I carried those rules.

Results

What a blind prior recovers

The answer is not yes or no (Table 2). The best blind rules recover 72% of the taxonomy ceiling in San Francisco while the weakest recover nothing in Austin. Had I tested one city I would have drawn a confident conclusion either way, and the city would have decided which.

The two intervals disagree. By ticket three of the four arms beat chance, but by category only the two San Francisco arms do, and Austin's model-written set spans [0.442, 0.770], which I cannot tell from guessing. The 72% itself spans roughly 7% to 93% on that unit, so read every claim below at that width.

What the rules add over structure alone

A rule set can score well because it is reasoning, or because almost any sensible cut of a lopsided taxonomy would. Two controls tell those apart.

Shuffling the verdicts across categories keeps everything

city	train	test	median resolution	categories
New York	72,206	78,296	5.4h	164
Chicago	17,291	38,671	118.0h	92
Austin	5,968	6,029	23.8h	117
San Francisco	17,723	17,007	15.1h	37

Table 1: the four feeds, two complete weeks from each.

city	rules	AUC	by ticket	by category	ceiling	recovered
Austin	mine	0.494	[0.481, 0.508]	[0.331, 0.649]	0.883	−1%
Austin	the model	0.611	[0.598, 0.626]	[0.442, 0.770]	0.883	29%
San Francisco	mine	0.676	[0.670, 0.684]	[0.515, 0.815]	0.886	46%
San Francisco	the model	0.777	[0.770, 0.784]	[0.529, 0.859]	0.886	72%

Table 2: what blind rules recover, by city and by author. Ticket intervals are 1,000 resamples, category intervals 2,000.

structural, the same taxonomy and row counts and the same set of scores, and breaks only which category received which score. That draws from the null directly, so the share of shuffles reaching the observed AUC is a permutation p-value (Table 3).

The null shuffles one verdict per category, so the arm has to be scored the same way for the two to be comparable. Scored per category it reaches 0.741 in San Francisco, and 2 of 2,000 shuffles match it, giving p of 0.0015. In Austin it reaches 0.617 and 220 shuffles match it, giving p of 0.110. So I reject the null in San Francisco and cannot in Austin. The reasoning does real work in one city and is not distinguishable from a lucky cut of the same taxonomy in the other.

Frequency is the harder baseline, because the pull gave me row counts and row counts are themselves real operational data. Scored rarer-is-slower, frequency alone reaches 0.672 in San Francisco against 0.676 for my own rules, so my rules add almost nothing beyond volume. The model’s rules clear frequency by 0.105. Picking that direction is one bit the catalogue did not give me, so the baseline is if anything generous.

The author effect is not established

The model beats me by 0.116 in Austin and 0.101 in San Francisco, the same direction and a similar size twice, which looks like something. By ticket both differences miss zero, at [0.104, 0.131] and [0.095, 0.107], but by category neither does, at [-0.036, 0.288] and [-0.021, 0.174]. Two cities are not enough to establish it, and an earlier draft of mine reported these differences with no interval at all.

What I predicted, before I looked

arm	predicted	observed	in band
Austin, mine	0.60 to 0.72	0.494	no
Austin, the model	0.52 to 0.62	0.611	yes
SF, mine	0.55 to 0.68	0.676	yes
SF, the model	none	0.777	-

I set the first band before the project had any result, and it was wrong. I set the next two after seeing Austin, so I already knew roughly what to expect, which is a weaker achievement and I count it as one. The fourth arm produces my headline 72% and carries no prediction at all. The only prediction I made in real ignorance is the one that failed.

One department explains most of the gap

Austin Resource Recovery (Table 4) is 34% of the city’s volume and 70% of its work runs slow, yet both sets of rules call it fast, at a mean prediction of 0.37 for the model and 0.05 for mine. Take it out and the model’s rules reach 70% against San Francisco’s 72%, while mine reach 38% against 46%, so the gap closes for both of us. The reason is plain enough. A missed collection sits open until the next scheduled route, a week away. The work is routine. The ticket is not, and nothing in the phrase ARR - Compost tells you so.

Run on San Francisco the same operation cuts the other way, and that is the more important half of the table. Its two largest categories, Street and Sidewalk Cleaning and Parking Enforcement, are 57% of test volume, and both are fast and both are called fast. Drop them and the ceiling barely moves, 0.886 to 0.851, while the model falls from 72% to 27% and I fall from 46% to 20%. A two-line rule calling those two fast and everything else slow scores 0.699 by itself. The 72% rests on a few high-volume decisions, not on 37 categories of reasoning.

Two further caveats. ARR is a department prefix rather than a category called waste, so removing it also removes fast work that department owns: ARR - Dead Animal Collection runs 0.0% slow and ARR - General 8.9%, both predicted fast and both right. And Graffiti Public is 8.2% of San Francisco volume and 85.1% slow but priced at 0.37 by the model and 0.09 by me, so the same failure ARR shows in Austin is present there and no subset here removes it.

What the prior actually beats

The sharpest way to put the San Francisco result is not as a percentage. Fitting the conditional rate on the real data using the category column alone scores 0.771, and the blind rules score 0.777. Rules written by an author who never saw an outcome beat a model that saw every outcome, given the same column. That is worth more than the 72%, because it needs no denominator. Two caveats: the rules also read the owning department, which the category-only conditional does not, so it is not quite like for like, and it does not hold in Austin, where the category-only conditional reaches 0.883 against the rules’ 0.611.

city	arm	AUC	[95% CI]	recovered	p vs H0
Austin	blind rules	0.611		29%	0.110
Austin	category frequency alone	0.516	[0.351, 0.726]	4%	
Austin	same rules, verdicts shuffled	0.501	[0.322, 0.674]	0%	
San Francisco	blind rules	0.777		72%	
San Francisco	category frequency alone	0.672	[0.441, 0.785]	45%	
San Francisco	same rules, verdicts shuffled	0.499	[0.312, 0.690]	0%	

Table 3: blind rules against two baselines built from structure alone, with category intervals and the permutation test of the null.

subset	rules	n	prior	ceiling	recovered
Austin, all	mine	6,029	0.494	0.883	−1%
Austin, all	the model	6,029	0.611	0.883	29%
Austin, minus ARR	mine	4,008	0.644	0.881	38%
Austin, minus ARR	the model	4,008	0.766	0.881	70%
SF, all	mine	17,007	0.676	0.886	46%
SF, all	the model	17,007	0.777	0.886	72%
SF, minus top 2	mine	7,255	0.571	0.851	20%
SF, minus top 2	the model	7,255	0.595	0.851	27%

Table 4: dropping the highest-volume categories in each city, for both sets of rules. The same operation run both ways.

The finding

Stated as an ordering rather than a level, because the level moves with which categories are in the subset and the interval on it is wide. Blind rules beat chance in San Francisco and not in Austin, the model’s rules beat mine in both cities, and removing one administratively clocked department closes most of the gap between the cities for both authors.

The explanation I offer for that ordering, that a blind prior holds where duration follows from the job and fails where the clock is an administrative cycle, is a hypothesis and not a result. I derived the partition from which categories my rules got wrong, in one city, so it cannot fail against the data that produced it. The test that would settle it is in the closing section.

What the real rates are worth

Rules written with a city’s rates in hand reach 81% there. That is above the 72% a blind author got in San Francisco, but not by much.

The last row is the one I did not expect. New York publishes the target it holds itself to, in days, for each complaint type. Ranking tickets by that alone reaches 0.830, or 79%, which matches rules written while looking at the real outcomes and needs no reasoning, no language model and no client records. If a client publishes what they intend to take, that document is worth about as much as the exercise this paper measures, and it is the first thing to ask them for.

It bears on contamination too, since a language model reasoning about municipal work may be recalling published material rather than reasoning, and this row shows such a document carries most of the signal. I could not find an equivalent dataset for Austin or San Francisco, the two cities the blind result rests on, which narrows that worry without settling it.

Carried to Chicago at Chicago’s own median they score 0.496, which I called actively misleading in an earlier draft. That does not hold up (Table 5). The two medians are 5.4h and 118.0h, twenty-two times apart, so the carried

row asks the rules a question nobody wrote them for. Held at New York’s own threshold the same rules reach 0.581, or 37%. What is left is mechanical, since the rules fire on only 17.1% of Chicago rows because they are New York acronyms.

Distribution fidelity misses all of this

My first recorded hypothesis, fixed in `HYPOTHESIS.md` before I ran anything, was that generated records matching real data to low-order fidelity would still hold combinations the real process could never produce, and that accuracy would fall as those rose. I tested it by fitting generators on real New York records, sampling, training on the sample and testing on real records [2]. It failed, for a structural reason. My pairwise generator draws department conditional on category, so it cannot emit a pair it never saw and its 0.0% impossible rate is an identity, and the learner is additive over one-hot columns with no interaction terms, so a wrong combination could not have hurt it either. Both were settled by the design before any data was touched. The work moved to the narrower question this paper answers, which is why my stated hypothesis and my pre-registration are not the same sentence.

The ladder still shows the one thing it is here for. Marginals alone lose almost everything, 0.511 against a 0.918 ceiling, while adding pairwise dependence recovers all of it at 0.917. A generator clearing the column shapes and column pair trends a quality report scores [5] is indistinguishable from real records here, so a distribution-level check cannot see the effect this paper measures. My generators are conditional probability tables with no privacy noise, so they are not PrivBayes or MST, but the order of structure they keep is the order those methods fit [3, 4].

Limits

Four cities, one domain, one task, and only two of the cities are blind. New York is a contamination control and Chicago its transfer target, so neither replicates anything,

rules	tested on	threshold	AUC	ceiling	recovered
New York	New York	5.4h	0.840	0.918	81%
NY rules carried	Chicago	118.0h	0.496	0.717	-2%
NY rules carried	Chicago	5.4h	0.581	0.717	37%
published target	New York	5.4h	0.830	0.918	79%

Table 5: rules written with the rates in hand, those rules carried to another city, and the city’s own published service target.

and the thing that does replicate is an author effect whose category intervals cross zero. A 311 feed also has three usable columns, no documents that have to agree with each other and no prices, so it is a thin stand-in for the enterprise data the question is really about.

The second author is a language model, and this is the biggest hole in my design. These feeds are among the most widely mirrored public datasets there are, so blind means the session showed it no durations, not that the weights hold none. Models have memorised popular tabular datasets and score better on the ones they have seen [11], which is the exact failure this arm is open to, and it produces every headline number. I did not record the prompt or the model version, so I cannot rerun it.

I wrote the San Francisco rules knowing what Austin had shown, so they are blind to that city’s rates but I am not blind to the lesson. The ARR result is a subset I chose after seeing which family my rules got wrong, so it fits the numbers and nothing has tested it. New York carries known artefacts too, including a department that closes most tickets at exactly midnight.

That fidelity and utility can come apart is established [6, 7, 8, 9], and whether the standard metrics measure the right thing is itself argued over [10]. I include the ladder only because it shows a distribution-level check cannot see the effect I measured.

The nearer literature is language models as priors over tabular problems, and this design is a small instance of it. Knauer et al. induce a decision tree from the target and the column names alone, with no rows and no labels [13], which is the move my model-written rules make, and TabLLM shows column names carry useful zero-shot signal before any label [14]. AutoElicit elicits a prior from a model and scores it against real data by marginal likelihood, choosing between priors by Bayes factor [15]. That does not depend on a test split, so it is a cleaner comparison than my ceiling share and is what I would adopt next time.

All four are classification where my target is duration, and the first three need feature names to be descriptive, which Knauer et al. state as a precondition. A 311 taxonomy is unusually kind on that point. An enterprise schema of coded columns would not be.

What it would take to answer the general question

Make the rules generate. My arms compare rule sets against a conditional fitted on real data, which measures how much of a conditional a person can guess, whereas the question asks what generated data costs. Sampling rows from the taxonomy and volume counts, labelling them with the prior, then training on those and testing

on real records would give the number a buyer needs: how much of the training value of real records does a generator built without them deliver.

Then predict the failures before looking. My ARR explanation says any category whose clock is an administrative cycle will break a blind prior. Name those categories in a fifth city from the taxonomy alone, write the list down, then measure. More authors would help too, including people who do the work, as would a model whose prompt and version I recorded.

One thing holds either way. A prior can only be evaluated as a prior once per dataset, before anyone has seen the outcomes, and the prediction has to come first. Of the predictions I made here, the only one made in real ignorance is the one that turned out wrong.

References

- [1] C. A. Field and A. H. Welsh. Bootstrapping clustered data. *J. R. Stat. Soc. B*, 69(3):369-390, 2007.
- [2] C. Esteban, S. L. Hyland and G. Rätsch. Real-valued (medical) time series generation with recurrent conditional GANs. arXiv:1706.02633, 2017.
- [3] J. Zhang, G. Cormode, C. M. Procopiuc et al. PrivBayes: private data release via Bayesian networks. *ACM Trans. Database Syst.*, 42(4):article 25, 2017.
- [4] R. McKenna, G. Miklau and D. Sheldon. Winning the NIST contest: a scalable and general approach to differentially private synthetic data. *J. Privacy and Confidentiality*, 11(3), 2021.
- [5] SDV Team. Quality report: column shapes and column pair trends. SDMetrics docs, docs.sdv.dev/sdmetrics, accessed 24 September 2026.
- [6] A. F. Karr, C. N. Kohnen, A. Oganian et al. A framework for evaluating the utility of data altered to protect confidentiality. *The American Statistician*, 60(3):224-232, 2006.
- [7] J. Snøke, G. M. Raab, B. Nowok et al. General and specific utility measures for synthetic data. *J. R. Stat. Soc. A*, 181(3):663-688, 2018.
- [8] L. Hansen, N. Seedat, M. van der Schaar et al. Reimagining synthetic tabular data generation through data-centric AI. In *NeurIPS Datasets and Benchmarks*, 2023.
- [9] B. van Breugel, Z. Qian and M. van der Schaar. Synthetic data, real errors: how (not) to publish and use synthetic data. In *ICML*, PMLR 202:34793-34808, 2023.
- [10] Y. Du and N. Li. Systematic assessment of tabular data synthesis. In *ACM CCS*, 2025.
- [11] S. Bordt, H. Nori, V. Rodrigues et al. Elephants never forget: memorization and learning of tabular data in large language models. In *COLM*, 2024.
- [12] A. C. Cameron, J. B. Gelbach and D. L. Miller. Bootstrap-based improvements for inference with clustered errors. *Rev. Econ. Stat.*, 90(3):414-427, 2008.
- [13] R. Knauer, M. Koddembrock, R. Wallsberger et al. Zero-shot decision tree induction and embedding with large language models. In *KDD*, pages 1196-1206, 2025.
- [14] S. Hegselmann, A. Buendia, H. Lang et al. TabLLM: few-shot classification of tabular data with large language models. In *AISTATS*, PMLR 206:5549-5581, 2023.
- [15] A. Capstick, R. G. Krishnan and P. Barnaghi. AutoElicit: LLMs for expert prior elicitation in predictive modelling. In *ICML*, PMLR vol. 267, 2025.